WORKED SOLUTION

STEP II 2016, Question 10

Mechanics • Hanging triangle

STEP II 2016, Question 10

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/16-s2-q10

The key idea

Only two things act on the triangle: its weight, at the centre of mass G, and the single contact force from the peg at the point of contact P. Two forces in equilibrium must be equal, opposite and along the same line, so the triangle swings round until G is directly below P. The question then becomes: how steeply must the peg's reaction lean away from the normal to BC, and is friction strong enough for that? So the first job is to locate G.

Solution

The model and the centre of mass

Treat the wire as three uniform rods with the same mass per unit length, and the peg as a single point of contact giving a normal reaction R (perpendicular to BC) and a friction force F (along BC), with $F \leq \mu R$.

Let $AB = AC = b$ and let M be the midpoint of BC. Since the angle at A is 2θ, the line AM bisects it and is perpendicular to BC, so

$$BM = MC = b\sin\theta, \qquad AM = b\cos\theta, \qquad BC = 2b\sin\theta.$$

By symmetry G lies on AM. Measure distances from BC towards A. The rods AB and AC each have mass proportional to b, with centres at their midpoints, at distance $\frac{1}{2}b\cos\theta$ from BC; the rod BC has mass proportional to $2b\sin\theta$ and its centre lies on BC. So the distance $d = GM$ is

$$d = \frac{b\cdot\frac{1}{2}b\cos\theta + b\cdot\frac{1}{2}b\cos\theta + 2b\sin\theta\cdot 0}{b+b+2b\sin\theta} = \frac{b\cos\theta}{2(1+\sin\theta)}.$$

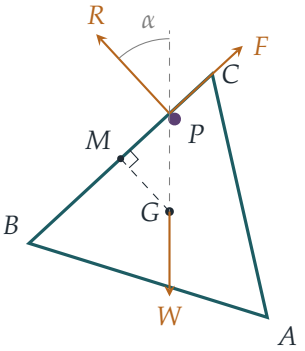
Equilibrium with the peg at a general point

Suppose the peg touches BC at P, with $MP = x$, where $0 \leq x \leq b\sin\theta$ (by symmetry we may take P between M and C). The triangle hangs below the peg, so the peg is

inside the triangle and BC rests on top of it.

Ask yourself The triangle can hang at any angle. How do we know which one it takes?

The forces on the triangle are the weight W at G and the peg's total contact force at P. With only two forces, equilibrium needs them to be equal and opposite (so the contact force is W vertically upwards) and to act along the same line (otherwise they would form a couple). So PG must be vertical: the triangle settles with G directly below P.



Let α be the angle between PG (the vertical) and the normal GM to BC. In the right-angled triangle PMG,

$$\tan\alpha = \frac{MP}{GM} = \frac{x}{d}.$$

The normal to BC makes angle α with the vertical, and BC makes angle α with the horizontal. Resolving perpendicular to and along BC:

$$R = W\cos\alpha, \qquad F = W\sin\alpha.$$

Both are non-negative since $0 \le \alpha < \frac{1}{2}\pi$, and $R > 0$. The triangle stays put without slipping provided $F \le \mu R$, that is,

$$\mu \ge \tan\alpha = \frac{x}{d}.$$

Every point of BC

The right-hand side x/d increases with x, so the condition is hardest to satisfy when P is at an end, $x = b\sin\theta$. Equilibrium is possible at every point of BC exactly when it is possible there:

$$\mu \ge \frac{b\sin\theta}{d} = b\sin\theta \cdot \frac{2(1+\sin\theta)}{b\cos\theta} = 2\tan\theta\,(1+\sin\theta),$$

which is the required condition. If it holds, then $\mu \ge x/d$ for every x in $[0, b\sin\theta]$, so the triangle can rest with the peg at any point of BC.

Checks

- At $x = 0$ (peg at M) we get $\alpha = 0$: the triangle hangs with BC horizontal and needs no friction at all, as symmetry says it should.
- With $\theta = 30^\circ$ the triangle is equilateral, $d = \frac{1}{3}b\cos 30^\circ$, and G should be the centroid, a third of the way up the height $b\cos 30^\circ$ from BC. It is. The condition becomes $\mu \geq 2 \cdot \frac{1}{\sqrt{3}} \cdot \frac{3}{2} = \sqrt{3} \approx 1.732$: at a corner, BC would have to sit at 60° to the horizontal.
- As $\theta \to 0$ the triangle becomes a thin spike, BC shrinks to nothing and the bound tends to 0; as $\theta \to \frac{1}{2}\pi$ it becomes a long flat strip and the bound blows up. Both fit intuition.
- A computer check, finding G by splitting the three rods into many small pieces and resolving forces at 201 points along BC for random θ and b, agreed with $2\tan\theta(1 + \sin\theta)$ to within 10^{-11}.

Common slips

- Using the centroid of the triangular *lamina* (a third of the way up AM) instead of the centre of mass of the *wire*. A wire frame has its mass on the sides, so BC counts with weight $2b\sin\theta$.
- Assuming BC stays horizontal and trying to take moments. The triangle tilts until G is below the peg; the tilt angle is what friction has to cope with.
- Checking only one point of BC. The worst case is the end of BC, and that is what fixes the bound.

Beyond STEP

Writing $\tan\alpha \leq \mu$ says that the total contact force must lie within the *cone of friction*, a cone of half-angle $\arctan\mu$ about the normal. This picture makes many friction problems (ladders, wedges, rough pegs) almost geometric, and it generalises to rigid bodies with several contacts, where the question becomes whether some choice of forces inside each cone can balance the load. Kleppner and Kolenkow, *An Introduction to Mechanics*, treat friction and rigid-body equilibrium with this kind of care.

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